

S1.1 Unit description

Mathematical models in probability and statistics; representation and summary of data; probability; correlation and regression; discrete random variables; discrete distributions; the Normal distribution.

S1.2 Assessment information

Examination

The examination will consist of one $1\frac{1}{2}$ hour paper. It will contain about seven questions of varying length. The mark allocations per question will be stated on the paper. All questions should be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Formulae which students are expected to know are given overleaf and these will not appear in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

This section lists formulae that students are expected to remember and that will not be included in formulae booklets.

Students will be expected to know and be able to recall and use the following formulae:

$$\text{Mean} = \bar{x} = \frac{\Sigma x}{n} \text{ or } \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{(\text{Variance})}$$

$$\text{Interquartile range} = \text{IQR} = Q_3 - Q_1$$

$$P(A') = 1 - P(A)$$

For independent events A and B ,

$$P(B|A) = P(B), P(A|B) = P(A),$$

$$P(A \cap B) = P(A) P(B)$$

$$E(aX + b) = aE(X) + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Cumulative distribution function for a discrete random variable:

$$F(x_0) = P(X \leq x_0) = \sum_{x \leq x_0} p(x).$$

$$\text{Standardised Normal Random Variable } Z = \frac{X - \mu}{\sigma}$$

$$\text{where } X \sim N(\mu, \sigma^2)$$
